Test2 Review, MAT 1575 Professor Chiu

1. Evaluate the following indefinite integrals:

$$(a)\int \sin^3(x)\,dx \qquad (b)\int \sin^5(x)\cos^2(x)\,dx \qquad (c)\int \sec^4(x)\,dx \qquad (d)\int \sin^2(x)\cos^2(x)\,dx$$

2. Evaluate the following indefinite integrals:

$$(a)\int \frac{dx}{\sqrt{1+9x^2}} \qquad (b)\int \frac{dx}{x^2\sqrt{1-x^2}} \qquad (c)\int \frac{\sqrt{x^2-9}}{x^4}dx$$

3. Evaluate the following indefinite integrals:

$$(a)\int \frac{2x^2+4x+12}{x^2+2x+5}dx \qquad (b)\int \frac{3x+2}{x^2+2x-8}dx \qquad (c)\int \frac{3x+4}{(x^2+4)(3-x)}dx$$

4. Evaluate the following improper integrals:

$$(a)\int_0^{\infty} \frac{5}{\sqrt[3]{x+5}}dx \qquad (b)\int_0^{1} \frac{\ln(x)}{\sqrt{x}}dx \qquad (c)\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$$

5.Find the Taylor polynomial of degree 2 for the given function, centered at the given number a:

$$(a) f(x) = e^x \ at\ a = 1 \qquad (b) f(x) = \cos(5x)\ at\ a = 2\pi \qquad (c) f(x) = \ln(x)\ at\ a = 1$$

6. For each of the following series, use the sequence of partial sums to determine whether the series converges or diverges

$$(a)\sum_{n=1}^{\infty} \frac{n}{n+2} \qquad (b)\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$$

Test2, MAT 1575 Professor Chiu

ID:_____ Name:_____

- Your grade: (Instructor Only.)

Total	1	2	3	4	5	6

- This test consists of 6 sets of questions for a total of 100 points.
- You have 60 minutes to complete the test.
- Wishing you success.

(10 points)1. True or False. Circle your answers (either T (true) or F (false)) on this sheet.

a)(T / F) $\int_{-1}^{5} \frac{dx}{x^3}$ is an improper integral.

b) (T / F) $\int_{1}^{\infty} \frac{dx}{x+1}$ is an improper integral.

c) (T / F) Given a sequence $\{a_n\} = \left\{\frac{n^{100}+3n^{50}+5}{5n^{100}+4n+1}\right\}$ for $n \geq 1$. Then this sequence converges to $\frac{1}{5}$.

d) (T / F) $\int_{2}^{5} \frac{dx}{x-1}$ is an improper integral.

e) (T / F) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \frac{x^8}{8!} + \cdots$

g) (T / F) Given a sequence $\{a_n\} = \left\{\left(1+\frac{1}{n}\right)^n\right\}$ for $n \geq 1$. Then this sequence converges to 1.

h)(T / F) *The harmonic series* $\sum_{n=1}^{\infty} \frac{1}{n}$ *is a convergent series.*

i)(T / F) $\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} = \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} + \frac{1}{5} - \frac{1}{6} + \frac{1}{6} - \frac{1}{7} + \cdots = \frac{1}{2}$

j)(T / F) Given a series $\sum_{n=1}^{\infty} n^2$, the sequence of partial sums of this series $\{S_n\}$ is $\{1, 4, 9, 16, 25, 36, 49, 64, \ldots\}$.

k)(T / F) Let $f(x) = \ln(x)$. Then the Maclaurin series of $f(x)$ exists.

From question 2. to 6., show all work and justify your answers.

(18 points) 2. Evaluate the following indefinite integrals:

$$(a) \int \sin^2(x)\cos^2(x)\,dx \qquad (b) \int \sec^4(x)\,dx$$

(18 points) 3. Evaluate the following indefinite integrals:

$$(a) \int \frac{3x+7}{x^2+6x+9}\,dx \qquad (b) \int \frac{2x^2+4x+12}{x^2+2x+5}\,dx \qquad (c) \int \frac{3x+2}{x^2+2x-8}\,dx$$

(18 points) 4. Evaluate the following improper integrals:

$$(a) \int_0^{\infty} \frac{3}{\sqrt[5]{x+3}}\,dx \qquad (b) \int_0^{1} \frac{dx}{\sqrt{1-x^2}}$$

Please turn over and finish the rest of the questions

(18 points) 5. Evaluate the following indefinite integrals:

$$(a)\int \frac{dx}{x^2\sqrt{1-x^2}} \qquad (b)\int \frac{\sqrt{x^2-9}}{x^4}dx$$

(18 points) 6.Find the Taylor polynomial of degree 4 for the given function, centered at the given number a:

$$(a) f(x) = e^{-2x} \text{ at } a = -1 \qquad (b) f(x) = \cos(2x) \text{ at } a = 2\pi$$

The end of the test.